

HORNSBY GIRLS' HIGH SCHOOL



MATHEMATICS YEAR 11 Preliminary Assessment Task 2 Half-Yearly Examination 2009

Student Number: _____

Time Allowed: 90 minutes plus 5 minutes reading time

Instructions:

- Attempt all questions
- Start a new page for each question
- The marks for each question are indicated
- Show all necessary working
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used

Outcomes Assessed:

- P1 Demonstrates confidence in obtaining realistic solutions to problems
P2 Provides reasoning to support conclusions in the correct context
P3 Performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
P4 Chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
P5 Understands the concept of a function and the relationship between a function and its graph

Marking Scheme:

	Total	
Algebra and Arithmetic	Q1.	/14
Functions and Graphs	Q2.	/14
Linear Functions	Q3.	/14
Trigonometric Functions	Q4.	/14
	Q5.	/14
Total Marks:		/70

This assessment task constitutes 30% of the final Preliminary Course Assessment.

QUESTION 1 (14 Marks) START A NEW PAGE**Marks**

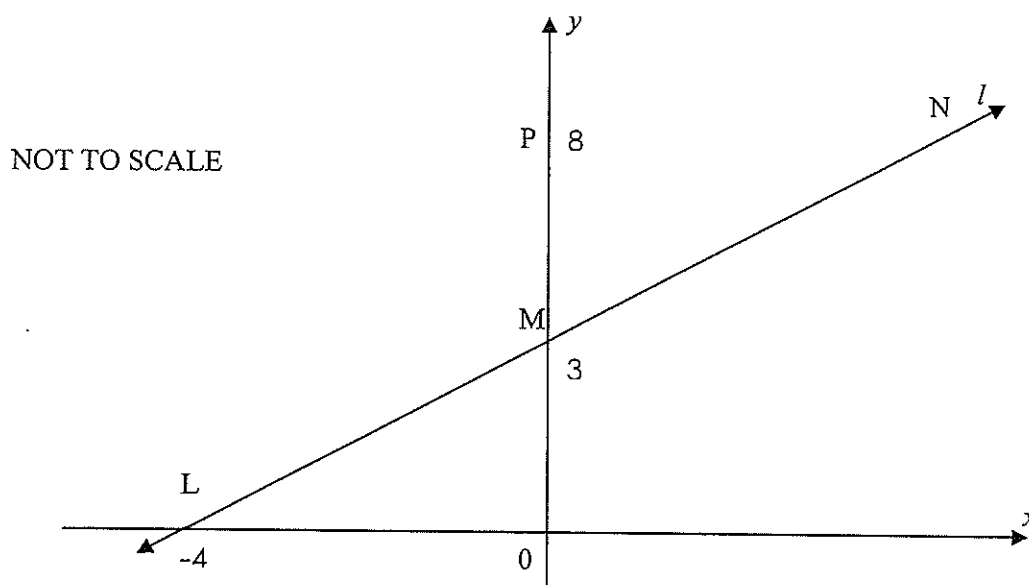
- a) Evaluate: $\sqrt{\frac{1.25 \times (3.6^4)}{0.075}}$ correct to 3 significant figures. 1
- b) Factorise: $3y^3 - 81$ 2
- c) Simplify: $(\sqrt{17} - 2\sqrt{5})(\sqrt{17} + 2\sqrt{5})$ 2
- d) Show that the only real solution of $|4x - 1| = 3 - 5x$ is $x = \frac{4}{9}$ 3
- e) If $f(x) = x^2 - 8x$, find $\frac{f(a+h) - f(a)}{h}$ 3
- f) Solve for x : $|7 - 2x| > 9$ and graph the solution on a number line. 3
-

QUESTION 2 (14 Marks) START A NEW PAGE**Marks**

- a) State the domain and range for (i) $y = \frac{8}{x-1}$ 2
- (ii) $y = \sqrt{16 - x^2}$ 2
- b) If $f(x) = 2^{-x} + 2^x$,
(i) find $f(a)$ and $f(-a)$. 1
(ii) hence, determine whether the function $f(x)$ is odd, even or neither. 1
- c) Sketch the following functions on separate number planes showing important features and intercepts. 8
- (i) $y = (x + 2)^2$
- (ii) $y = |x + 2|$
- (iii) $y = \frac{1}{x+2}$
- (iv) $y = 2 - x^3$

QUESTION 3 (14 Marks) **START A NEW PAGE****Marks**

a)



The line l cuts the x axis at $L (-4,0)$ and the y axis at $M (0,3)$. N is a point on the line l and P is the point $(0,8)$

- | | | |
|-------|--|---|
| (i) | Find the equation of the line l in general form. | 2 |
| (ii) | Show that $\triangle LMP$ is isosceles. | 2 |
| (iii) | M is the midpoint of LN find the coordinates of N . | 2 |
| (iv) | Show that $\angle NPL$ is a right angle. | 2 |
| (v) | Find the perpendicular distance from the point $Q(-4,5)$ to the line l . | 2 |
| (vi) | Find the gradient of PQ and hence determine if PQ is parallel to line l , LN . | 2 |
- b) Find the equation of the straight line that makes an angle of 135° with the positive direction of x axis and passes through the point $(2,-3)$. 2

QUESTION 4

(14 Marks)

START A NEW PAGE

Marks

a) Simplify (i) $\frac{1}{\sec(-\theta)} = \frac{1}{\sec \theta} = \cos \theta$

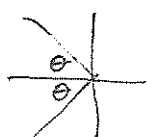
(ii) $\frac{\csc(180^\circ - \theta) \cdot \sin(90^\circ - \theta)}{\csc \theta - \cos \theta} = \cot \theta$

(iii) $2 - 2\sin^2 \theta = 2(1 - \sin^2 \theta) = 2\cos^2 \theta$

1
2
1
4

b) Solve $\cos \theta = -0.6$ for $0^\circ \leq \theta \leq 360^\circ$. Give your answer to the nearest minute.

2
5



$\theta = 126^\circ 52'$
 $233^\circ 8'$

c) Evaluate, giving the exact value as a single fraction for

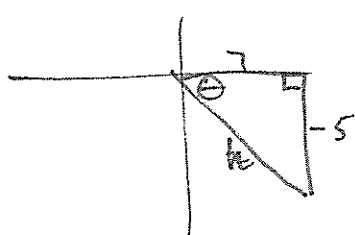
(i) $\tan 210^\circ = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}$

(ii) $\sin 60^\circ + \tan 210^\circ = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} = \frac{5\sqrt{3}}{6}$

1
2

d) Given that $\tan \theta = \frac{-5}{7}$ and $\cos \theta > 0$, find the exact value of $\csc \theta$. Show necessary working.

3
6



$h^2 = 49 + 25$
 $= 74$
 $h = \sqrt{74}$

$\therefore \csc \theta = \frac{\sqrt{74}}{-5}$

e) Prove that $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$

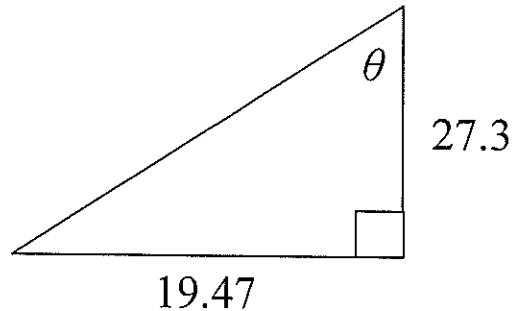
3

LHS = $\frac{1 - \tan^2 \theta}{\sec^2 \theta}$
 $= \frac{1 - \frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{1} = \cos^2 \theta - \sin^2 \theta = \text{RHS}$

QUESTION 5 (14 Marks)**START A NEW PAGE****Marks**

- a) Find θ to the nearest minute.

2



- b) A ship leaves port and sails 58 kilometres on a bearing of 138° . How far due south of the port is the ship? Draw a diagram and give your answer correct to the nearest kilometre.

2

- c) From an aircraft flying at 9 000m above sea level the angle of depression of a ship is $33^\circ 25'$. How far is the ship from the aircraft in a straight line? (Give your answer correct to nearest metre).

2

- d) Solve for $0^\circ \leq \theta \leq 360^\circ$

(i) $\operatorname{cosec} \theta = 2$

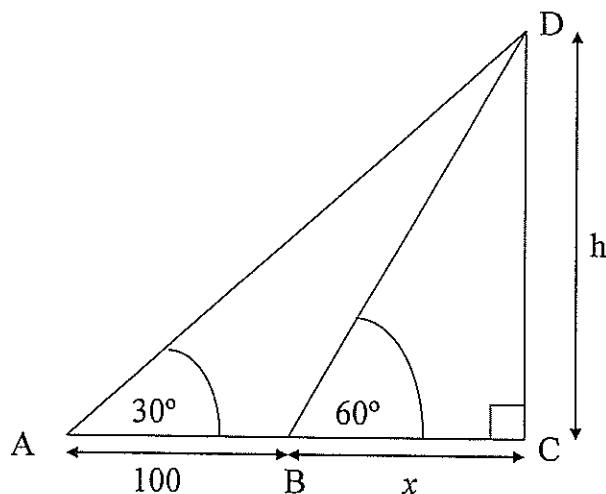
2

(ii) $\cot \theta = 0$

2

- e) A child at A observes the angle of elevation of a kite caught in a tree to be 30° . She then walks 100 metres towards the kite to a point B and finds the angle of elevation to be 60° . If the kite has a height of h metres and the child has x metres still to walk before she is directly under the kite, find x .

4

**END OF EXAMINATION**

Q1 a) $\div 52.9089 \dots$ (from calculator)
 $= 52.9$ (3 significant figures) | 1

b) $3(y^3 - 27) = 3(y-3)(y^2 + 3y + 9)$ | 2

c) $17 - 20 = -3$ | 2

d) $4x - 1 = 3 - 5x$ OR $-(4x - 1) = 3 - 5x$
 either $9x = 4$ $-4x + 1 = 3 - 5x$
 $x = \frac{4}{9}$ $x = 2$

Check: $|4 \times \frac{4}{9} - 1|$
 $LHS = \frac{7}{9}$

$RHS = 3 - 5 \times \frac{4}{9}$
 $= \frac{7}{9}$

$LHS = RHS$
 $\therefore x = \frac{4}{9}$ is the only solution.

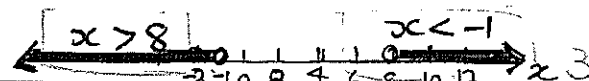
Check: $|4 \times 2 - 1|$
 $LHS = 7$
 $RHS = 3 - 5 \times 2 = -7$

$\therefore$ Not a solution
 $LHS \neq RHS$

e) $f(a+h) = (a+h)^2 - 8(a+h)$
 $= a^2 + 2ah + h^2 - 8a - 8h$

$f(a) = a^2 - 8a$

$\therefore \frac{f(a+h) - f(a)}{h} = \frac{a^2 + 2ah + h^2 - 8a - 8h - (a^2 - 8a)}{h}$
 $= \frac{2ah + h^2 - 8h}{h}$
 $= \frac{h(2a + h - 8)}{h}$
 $= 2a + h - 8$ | 3

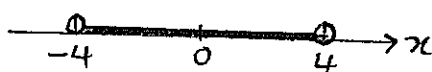
f) $|7 - 2x| > 9$
 either $-(7 - 2x) > 9$ OR $7 - 2x > 9$
 $7 - 2x < -9$ OR $-2x > 2$
 $-2x < -16$ $-2x > 2$
 $x > 8$ $x < -1$
 | 3

Q2. a) i) D: $x \neq 1$

R: $y \neq 0$

(ii) D: $16 - x^2 \geq 0$

$(4-x)(4+x) \geq 0$



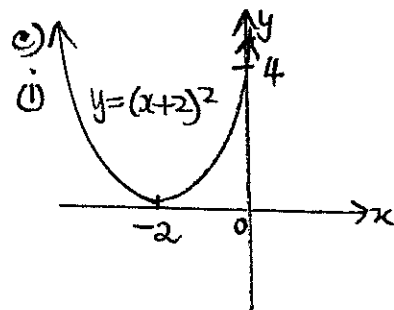
Test $x = 0$ $16 \geq 0$ ✓

$\therefore -4 \leq x \leq 4$

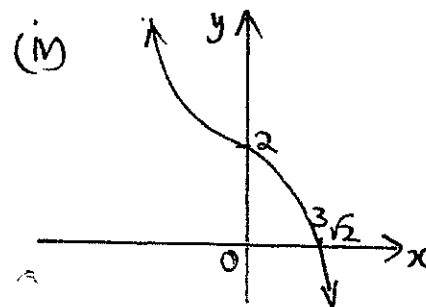
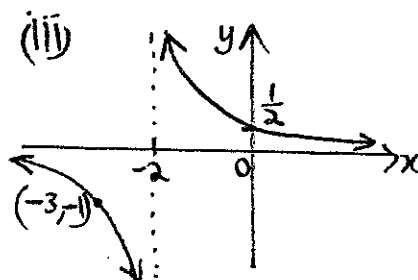
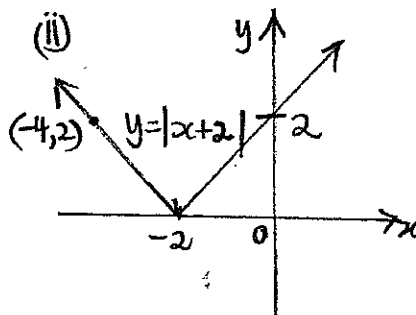
R: $y \geq 0$ | 4

Q2 b) $f(a) = 2^{-a} + 2^a$
 $f(-a) = 2^a + 2^{-a}$

$\therefore f(-a) = f(a) \therefore$ Function is even. | 2



2 each $\left\{ \begin{array}{l} \text{shape} \\ \text{intercepts} \end{array} \right.$



Q3. a) i) $\frac{x}{-4} + \frac{y}{3} = 1$

$3x - 4y = 12$

$3x - 4y + 12 = 0$

ii) PM = 5 units

LM = 5 units {3, 4, 5} Pythagorean Triad

or $d = \sqrt{3^2 + 4^2}$
 $= \sqrt{25}$
 $= 5$

$\therefore \triangle LMP$ is isosceles (2 equal sides)

iii) $(0, 3) = \left(\frac{-4+x_1}{2}, \frac{0+y_1}{2} \right)$

$-4+x_1 = 0$, $\frac{0+y_1}{2} = 3$
 $x_1 = 4$, $y_1 = 6$

$\therefore N$ is $(4, 6)$

iv) $m_{NP} = \frac{8-6}{0-4}$, $m_{PL} = \frac{8-0}{0+4}$
 $= -\frac{1}{2}$, $= 2$

$\therefore m_{NP} \times m_{PL} = -\frac{1}{2} \times 2$
 $= -1$

$\therefore NP \perp PL$

$\therefore \angle NPL$ is a right angle.

v) $d = \frac{|3(-4) - 4(5) + 12|}{\sqrt{3^2 + 4^2}}$
 $= \frac{|-20|}{5}$ units.
 $= 4$

vi) $m_{PQ} = \frac{8-5}{0+4}$, $m_{LN} = \frac{3}{4}$
 $= \frac{3}{4}$ $\therefore$ gradients are equal
 $\therefore PQ \parallel LN$ (line l)

b) $m = \tan 135^\circ$
 $= -1$

$y = -1x + b$

$-3 = -1 \times 2 + b$

$b = -1$

$\therefore y = -x - 1$

Q4. a) i) $\cos(-\theta) = \cos \theta$

ii) $\frac{1}{\sin(180^\circ - \theta)} \times \frac{\cos \theta}{\sin \theta} = \frac{\cos \theta}{\sin \theta}$
 $= \cot \theta$

iii) $2(1 - \sin^2 \theta) = 2 \cos^2 \theta$

b) Quadrants II, III $\cos \theta < 0$

Acute angle α is $53^\circ 7' 48''$ (from calculator)

$\therefore \theta = 180^\circ - \alpha$ or $\theta = 180^\circ + \alpha$
 $\div 126^\circ 52' 12''$ $\div 233^\circ 7' 48''$

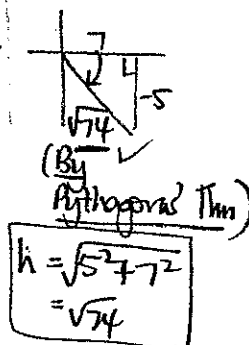
$= 126^\circ 52'$ (correct nearest minute) $= 233^\circ 8'$ (correct nearest minute)

c) i) $\tan 210^\circ = \tan(180^\circ + 30^\circ)$ 3rd quadrant $\tan \theta > 0$
 $= \tan 30^\circ$
 $= \frac{1}{\sqrt{3}}$

ii) $\sin 60^\circ + \tan 210^\circ = \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}}$
 $= \frac{3+2}{2\sqrt{3}}$
 $= \frac{5}{2\sqrt{3}}$ or $\frac{5\sqrt{3}}{6}$

d) $\left. \begin{matrix} \tan \theta < 0 \\ \cos \theta > 0 \end{matrix} \right\} \therefore$ 4th quadrant.
 $\therefore \sin \theta < 0$

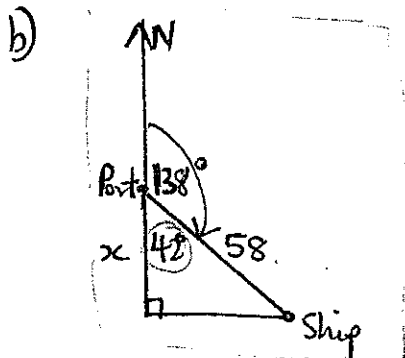
$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$
 $= \frac{1}{-\frac{5}{\sqrt{74}}}$
 $= -\frac{\sqrt{74}}{5}$


 $h = \sqrt{5^2 - 7^2}$
 $= \sqrt{74}$

e) LHS = $\left(1 - \frac{\sin^2 \theta}{\cos^2 \theta} \right) \div \sec^2 \theta$
 $= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\sec^2 \theta}$
 $= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \times \cos^2 \theta$
 $= \text{RHS.}$

Q5. a) $\tan \theta = \frac{19.47}{27.3}$

$\theta \doteq 35^\circ 29' 45''$ (from calc.)
 $= 35^\circ 30'$ (to nearest minute) 2

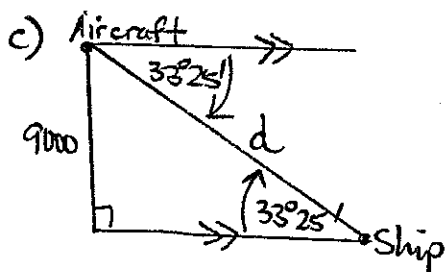


$\cos 42^\circ = \frac{x}{58}$

$x = 58 \cos 42^\circ$

$\doteq 43.10...$ (from calculator)

$\therefore$ Ship is 43 km south of the Port 2



$\frac{9000}{d} = \sin 33^\circ 25'$

$d = \frac{9000}{\sin 33^\circ 25'}$

$\doteq 16342.13...$ (from calculator) 2

$\therefore$ The distance between them is 16 342 m.

d) i) $\operatorname{cosec} \theta = 2, 0^\circ \leq \theta \leq 360^\circ$

$\therefore \frac{1}{\sin \theta} = 2$

$\sin \theta = \frac{1}{2}$ $\sin \theta > 0$
 $\therefore \text{I, II quadrants}$

$\theta = 30^\circ, 180^\circ - 30^\circ$

$= 30^\circ, 150^\circ$ 2

Q5d) (i) $\cot \theta = 0, 0^\circ \leq \theta \leq 360^\circ$

$\frac{1}{\tan \theta} = 0$

$\tan \theta$ is undefined.

$\therefore \theta = 90^\circ, 270^\circ$ 2

e) $\frac{h}{x} = \tan 60^\circ$
 $h = x \tan 60^\circ$

$\frac{h}{(100+x)} = \tan 30^\circ$

$h = (100+x) \tan 30^\circ$

$x \tan 60^\circ = 100 \tan 30^\circ + x \tan 30^\circ$

$x \tan 60^\circ - x \tan 30^\circ = 100 \tan 30^\circ$

$x (\tan 60^\circ - \tan 30^\circ) = 100 \tan 30^\circ$

$x = \frac{100 \tan 30^\circ}{\tan 60^\circ - \tan 30^\circ}$
 $= \frac{100 \times \frac{1}{\sqrt{3}}}{\sqrt{3} - \frac{1}{\sqrt{3}}}$

$= \frac{100}{\sqrt{3}} \div \left(\frac{3-1}{\sqrt{3}} \right)$

$= \frac{100}{\sqrt{3}} \times \frac{\sqrt{3}}{2}$

$= 50$ 1

